HSA 13 and TDS Missions

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I. ATMOSPHERIC COUPLING AND IONOSPHERIC LIFT TENSORS

The departure from Earth ($T + 0$) utilizes the **Magneto-Aerodynamic (MA) Interface**. The HSA-13 utilizes an **Active Ionizing Forebody** to mitigate the stagnation temperature T_s. At $Mach \approx 5$, the forward ring initiates a 14.28 GHz pulse, transforming the local atmosphere into a controllable plasma. The fluid-magnetic interaction is governed by the **Hartmann Number (Ha)**:

$$Ha = BL\sqrt{\frac{\sigma}{\eta}} \tag{1}$$

Where $B = 0.6$ T, $L = 12$ m, σ is electrical conductivity, and η is dynamic viscosity. By maintaining $Ha > 100$, magnetic forces dominate the fluid dynamics. The **Navier-Stokes Equation** is modified with the **Lorentz Force term**:

$$\rho\left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v}\right) = -\nabla p + \mu \nabla^2 \mathbf{v} + (\mathbf{J} \times \mathbf{B}) \tag{2}$$

This creates a **Plasma Bow Shock** with a standoff distance $\Delta \approx 1.5$ m, diverting thermal flux. Steering is achieved through the **Asymmetric Field Gradient (∇B)**, yielding a lift-to-drag ratio (L/D) of 4.5:1 in the mesosphere.

II. THE TDS PATHFINDER AND STOCHASTIC RANGING LOGIC

The **TDS Master Unit** (100 kg) acts as the mathematical anchor. At $V \approx 1,271,376$ m/s, traditional telemetry fails. The TDS calculates position via the **Phase-Shift of Solar Background Noise (ξ)**:

$$\Delta x = \frac{c}{2\pi f} \cdot \arccos(\mathrm{Corr}(\xi_{sun}, \xi_{local})) \tag{3}$$

With a 15g initial acceleration, the TDS maps the **Interplanetary Magnetic Field (IMF)** to calibrate the AV-E Funnel for optimal proton density.

III. DEEP-SPACE PROPULSION: ACTIVE VACUUM ENTRAINMENT(AV-E)

The engine harvests solar wind as reaction mass. The effective capture area (A_{eff}) is a magnetic dipole funnel with a radius determined by the pressure equilibrium:

$$R_{funnel} = \left(\frac{\mu_0 \mu^2}{4\pi^2 \rho v^2}\right)^{1/6} \tag{4}$$

At $0.0042c$, $R \approx 250$ km. The mass-flow rate ($\dot{m}$) of protons is defined by:

$$\dot{m} = \rho_{plasma} \cdot A_{eff} \cdot V_{ship} \approx 4.2 \times 10^{-4} \text{ kg/s} \tag{5}$$

Acceleration is achieved via the **Lorentz Acceleration Vector** in the Nb3Sn Superconducting Toroid:

$$\mathbf{a}_{plasma} = \frac{1}{m}[q(\mathbf{E} + \mathbf{v} \times \mathbf{B})] \tag{6}$$

Resulting in an exhaust velocity $V_e = 2,450$ km/s and a constant thrust $F = 29,430$ N, sustaining $1g$ for the 72-hour transit.

IV. THERMAL ERASURE AND THE LANDAUER LIMIT

The HSA-13 treats incident heat as **Entropy (S)**. Based on **Landauer's Principle**, we use external heat to perform the computational work of maintaining system order. The **Entropy Flux Equation** is:

$$\frac{dS_{total}}{dt} = \frac{\dot{Q}}{T} + \dot{S}_{gen} \leq 0 \tag{7}$$

By forcing the **Stationary Social Equilibrium ($x^* = 0.842$)**, the ship acts as a **Negative Entropy Sink**. Thermal energy (10^{15} W) is converted into the ordering work of the magnetic field lattice:

$$W_{order} = \int k_B T \ln(2)\, dI \tag{8}$$

V. THE 72-HOUR FLIP AND MARTIAN BRAKING

At $T + 36$h, after $82,386,000$ km, the craft rotates $180°$. The MHD-Drive operates in reverse bias, utilizing the **Martian Magnetosphere** and IMF for deceleration:

$$\Delta V = \int_{36h}^{72h} \frac{\mathbf{J} \times \mathbf{B}}{M_{ship}}\, dt \tag{9}$$

With $M_{ship} = 3,000$ kg (dry mass), the deceleration curve is perfectly symmetrical. Arrival precision is $\Delta x \approx 4.2$ m.

VI. FINAL NUMERICAL VALIDATION

Metric	**Value**	**Validation**
Specific Impulse (I_{sp})	∞	AV-E Harvesting
Peak Velocity (V_{max})	1,271,376 m/s	1g Constant Accel
Launch Integration	Nov 15, 2026	TDS Pathfinder
Launch Integration	April 1, 2027	HSA-13

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Approved by Heidenbillg Board: ______________________

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